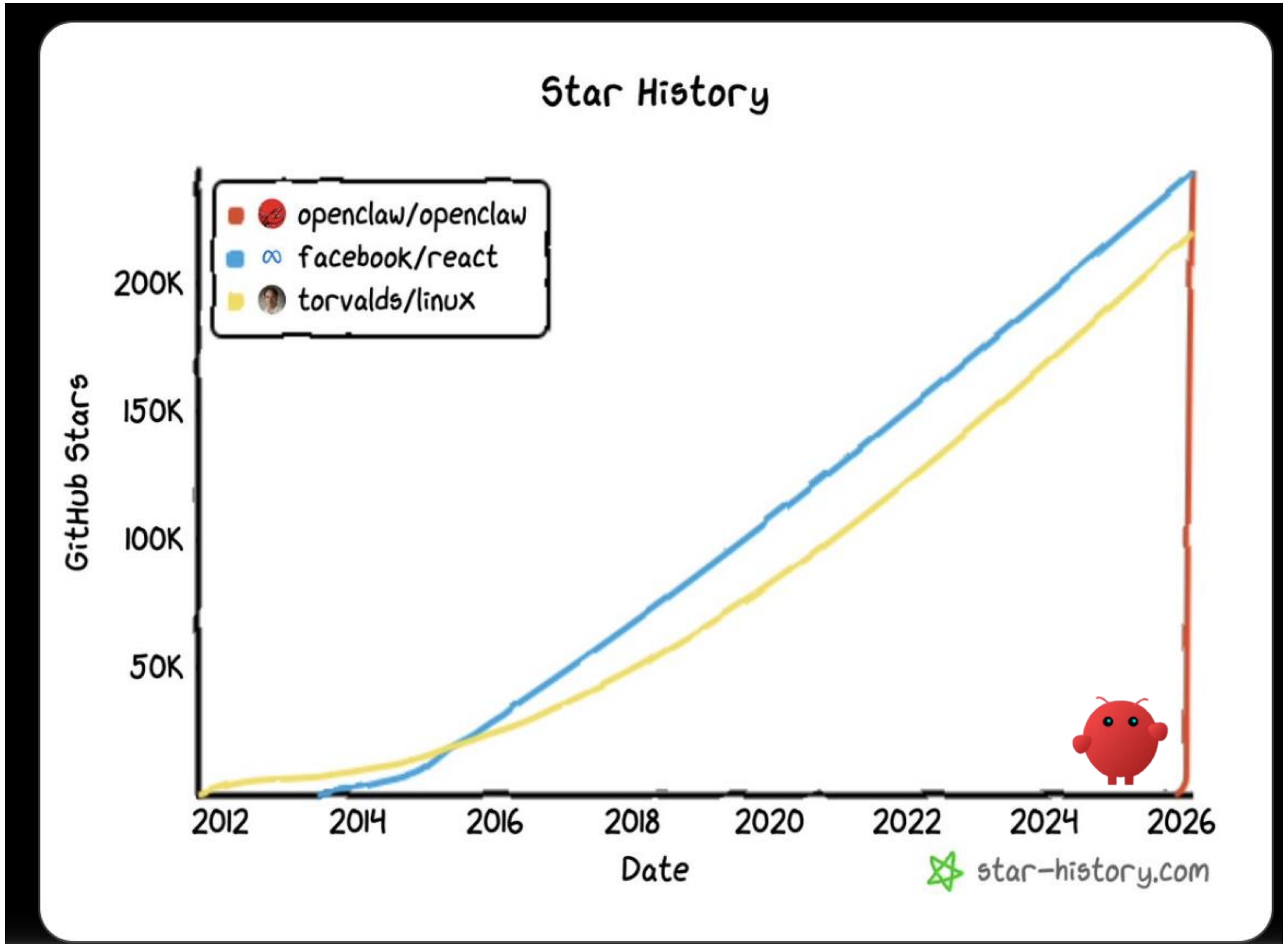
The Era of Agentic AI

Dr. Charles Cheung

OpenClaw – the ChatGPT Moment for Autonomous, Self-Evolving Agents



The Evolution of AI Agents

From LLMs and multi-turn prompts to autonomous, self-evolving agents working on your behalf

ChatGPT
Nov 2022

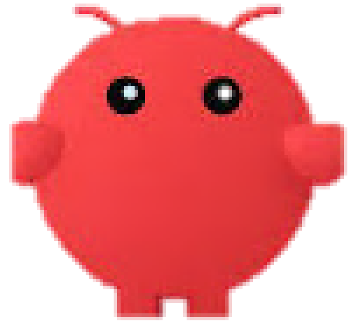
Foundation LLMs,
Multi-turn Prompts

DeepSeek
Jan 2025

Reasoning Translates
to Better Outcomes

OpenClaw
Jan 2026

Autonomous, Self-Evolving
Agents



“we’re going to need a lot more compute where we’re going”

Andrej Karpathy
@karpathy

haha actually - huge successover the weekend, i just wanted to write it up a bit but didn't get enough time to finish that. many people's reaction was why would you need so much compute(mac mini) but i found it's not enough compute, even after addin my DGX spark to the home.compute fabric. we're going to need a lot more compute where we're going

6:41 PM · Feb 23, 2026 · **452.3K** Views

NVIDIA OpenShell

Safely Build Autonomous, Self-Evolving Agents

Sandbox

Isolated execution environment for long running agents

Policy Engine

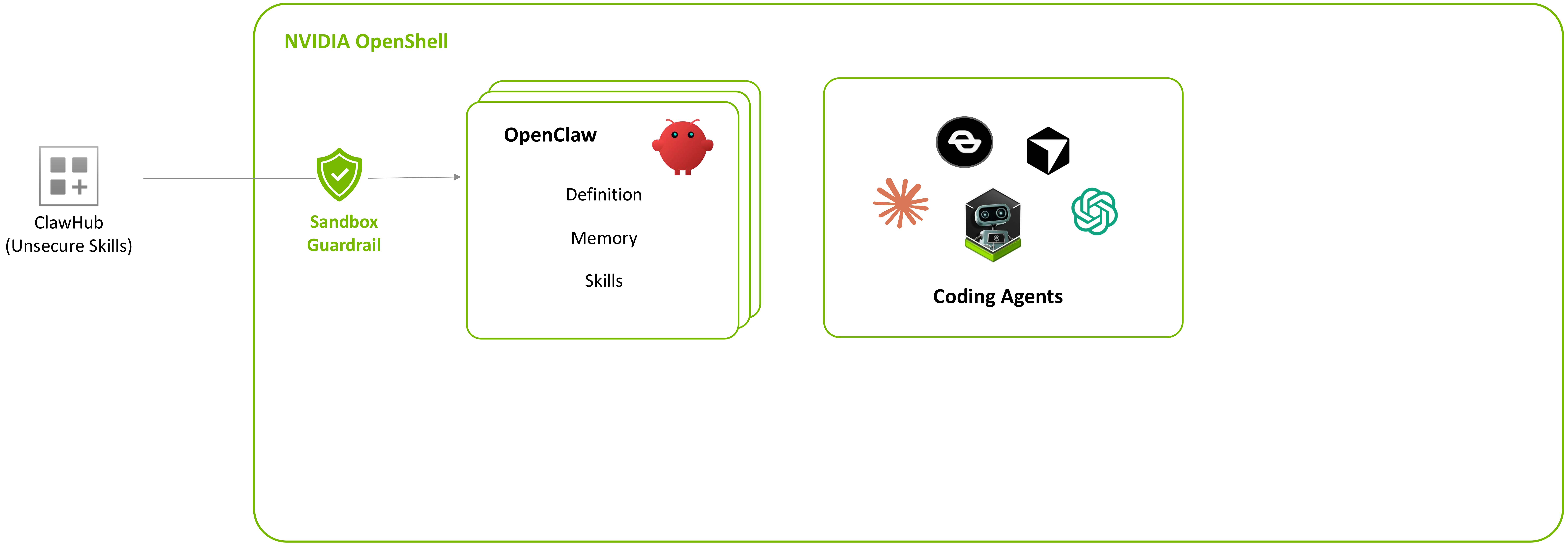
Granular oversight on installing packages and spawning sub-agents

Privacy Router

Local inference for private tasks and network access for others

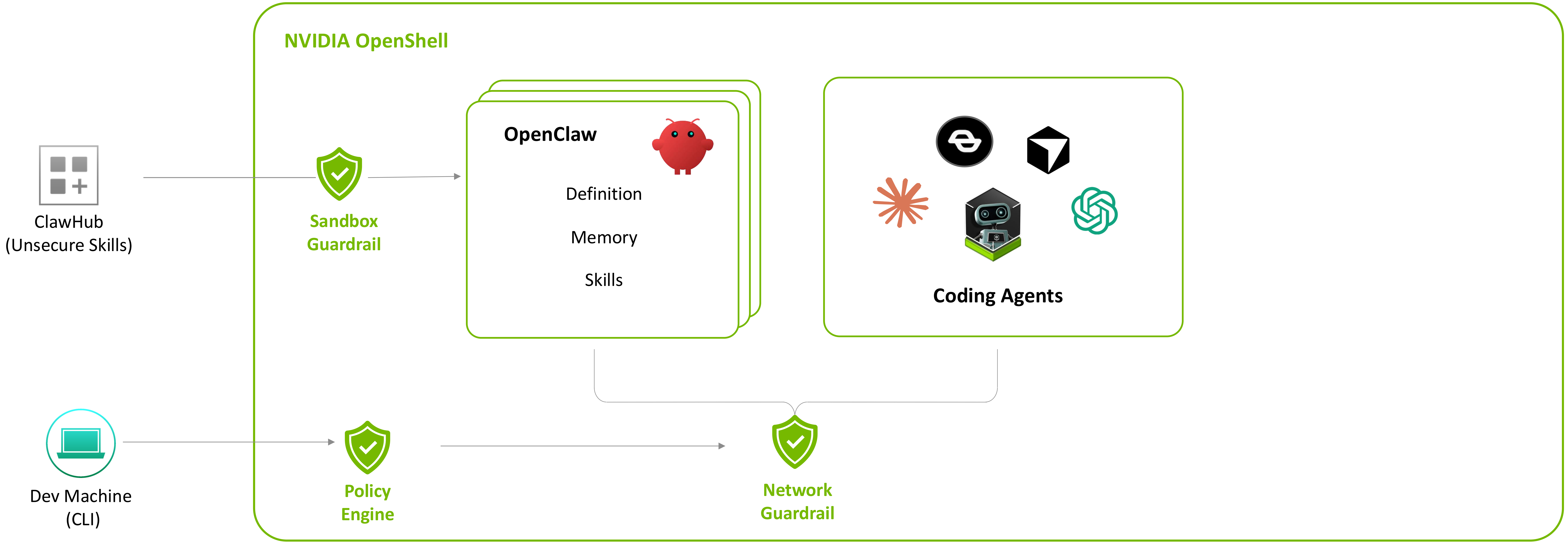
NVIDIA OpenShell - The Safer Runtime for Agents

Safely Build and Run Autonomous, Self Evolving Agents



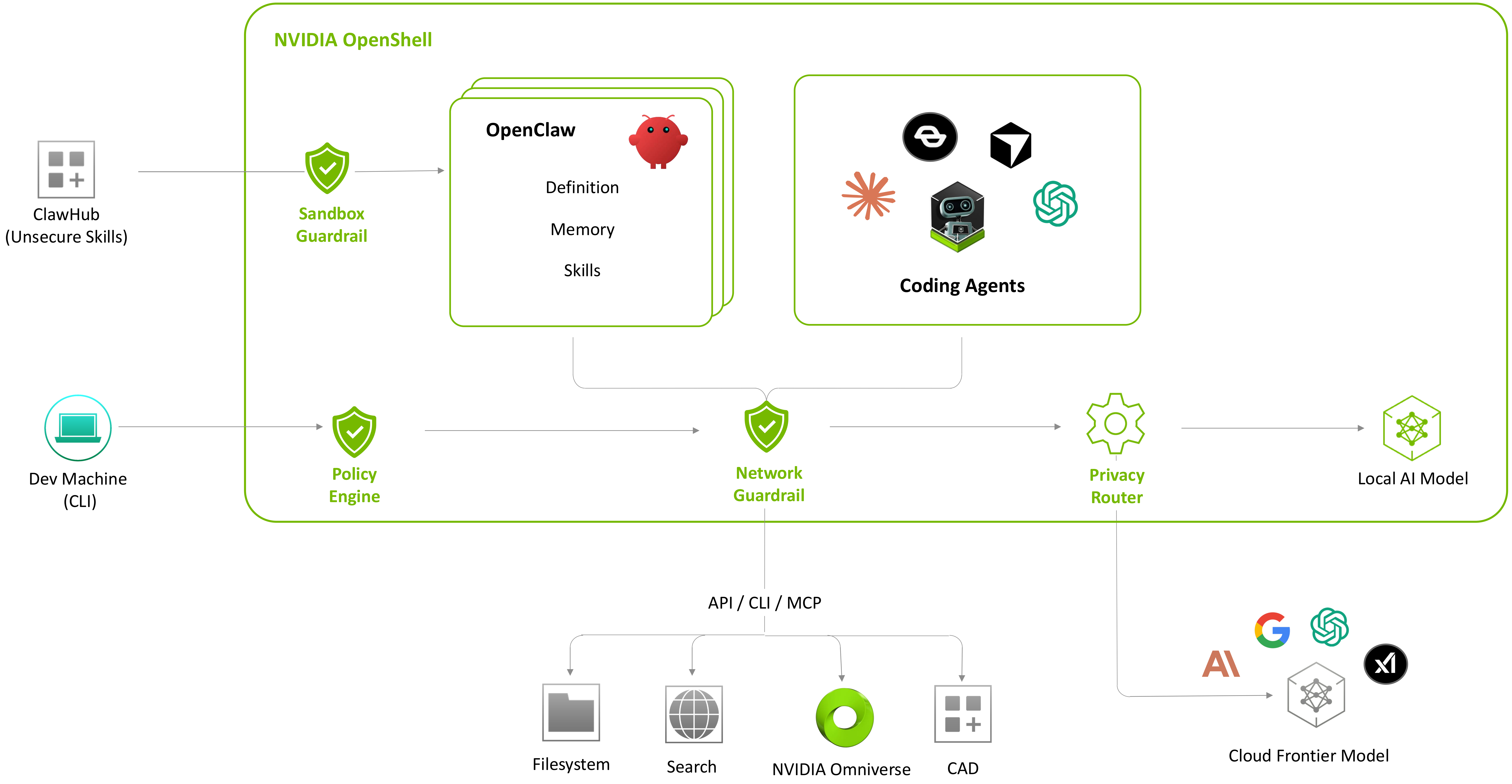
NVIDIA OpenShell - The Safer Runtime for Agents

Safely Build and Run Autonomous, Self Evolving Agents



NVIDIA OpenShell - The Safer Runtime for Agents

Safely Build and Run Autonomous, Self Evolving Agents



Common Risks and Controls

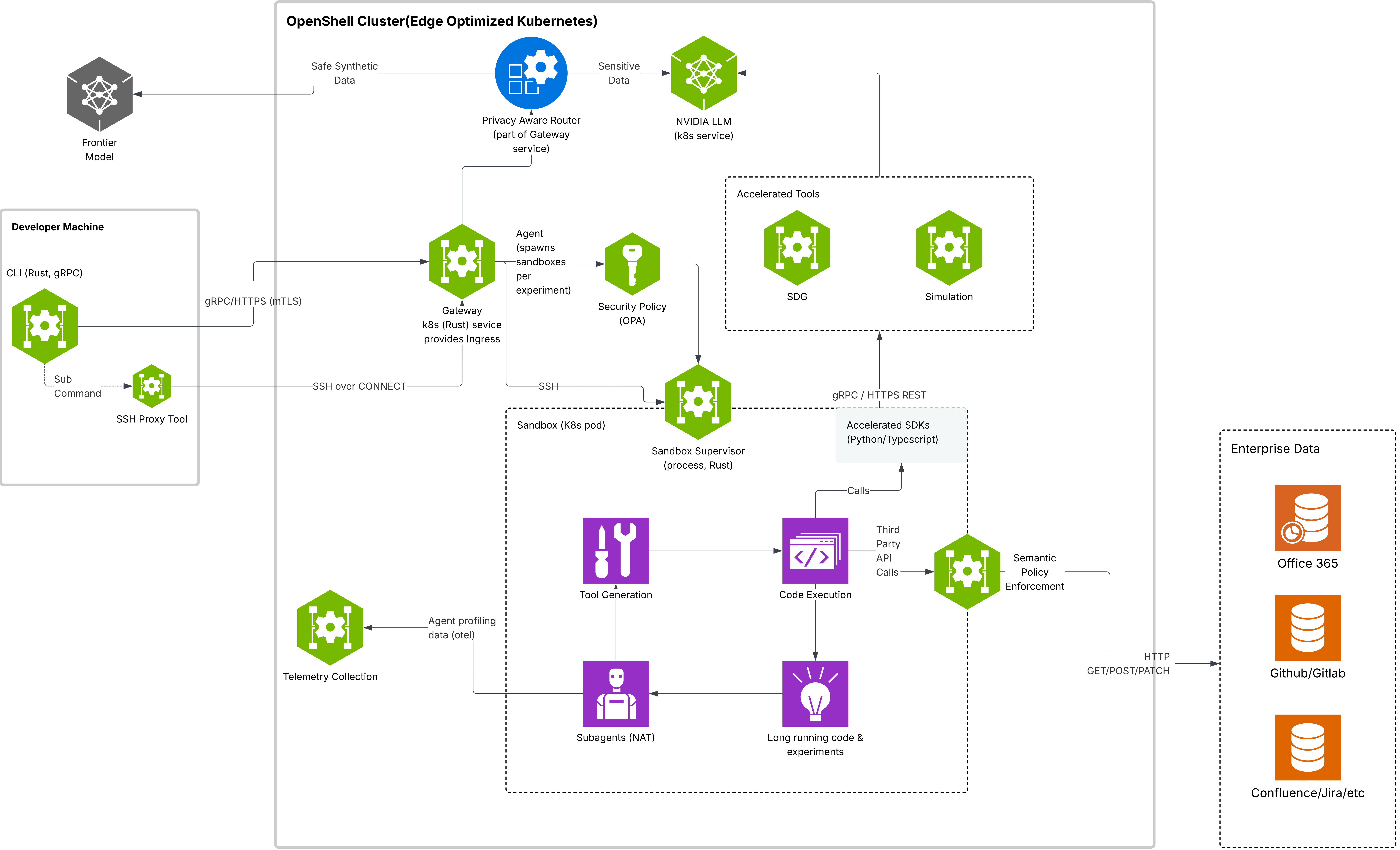
Threat	Without controls	With OpenShell
Data exfiltration	Agent uploads source code or internal files to unauthorized endpoints.	Network policies allow only approved destinations; other outbound traffic is denied.
Credential theft	Agent reads local secrets such as SSH keys or cloud credentials.	Filesystem restrictions (Landlock) confine access to declared paths only.
Unauthorized API usage	Agent sends prompts or data to unapproved model providers.	Privacy routing and network policies control where inference traffic can go.
Privilege escalation	Agent attempts `sudo`, setuid paths, or dangerous syscall behavior.	Unprivileged process identity and seccomp restrictions block escalation paths.

Protection Layers

OpenShell applies defense in depth across the following policy domains.

Layer	What it protects	When it applies
Filesystem	Prevents reads/writes outside allowed paths.	Locked at sandbox creation.
Network	Blocks unauthorized outbound connections.	Hot-reloadable at runtime.
Process	Blocks privilege escalation and dangerous syscalls.	Locked at sandbox creation.
Inference	Reroutes model API calls to controlled backends.	Hot-reloadable at runtime.

How it works



OpenShell Cluster(Edge Optimized Kubernetes)
Safe Synthetic Data
Sensitive Data
Frontier Model
Privacy Aware Router (part of Gateway service)
NVIDIA LLM (k8s service)
Developer Machine
CLI (Rust, gRPC)
gRPC/HTTPS (mTLS)
Gateway k8s (Rust) sevice provides Ingress
Agent (spawns sandboxes per experiment)
Security Policy (OPA)
Accelerated Tools
SDG
Simulation
Sub Command
SSH Proxy Tool
SSH over CONNECT
SSH
gRPC / HTTPS REST
Sandbox (K8s pod)
Sandbox Supervisor (process, Rust)
Accelerated SDKs (Python/Typescript)
Calls
Tool Generation
Code Execution
Third Party API Calls
Semantic Policy Enforcement
HTTP GET/POST/PATCH
Enterprise Data
Office 365
Github/Gitlab
Confluence/Jira/etc
Telemetry Collection
Agent profiling data (otel)
Subagents (NAT)
Long running code & experiments

Components

Component	Role
Gateway	Control-plane API that coordinates sandbox lifecycle and state, acts as the auth boundary, and brokers requests across the platform.
Sandbox	Isolated runtime that includes container supervision and policy-enforced egress routing.
Policy Engine	Policy definition and enforcement layer for filesystem, network, and process constraints. Defense in depth enforces policies from the application layer down to infrastructure and kernel layers.
Privacy Router	Privacy-aware LLM routing layer that keeps sensitive context on sandbox compute and routes based on cost and privacy policy.

Deployment Modes

Mode	Description	Command
Local	The gateway runs inside Docker on your workstation. The CLI provisions it automatically on first use.	`openshell gateway start`
Remote	The gateway runs on a remote host via SSH. Only Docker is required on the remote machine.	`openshell gateway start --remote user@host`
Cloud	A gateway already running behind a reverse proxy (e.g. Cloudflare Access). Register and authenticate via browser.	`openshell gateway add https://gateway.example.com`

Supported Agents

Agent	Source	Default Policy	Notes
Claude Code	`base`	Full coverage	Works out of the box. Requires `ANTHROPIC_API_KEY`.
OpenCode	`base`	Partial coverage	Pre-installed. Add `opencode.ai` endpoint and OpenCode binary paths to the policy for full functionality.
Codex	`base`	No coverage	Pre-installed. Requires a custom policy with OpenAI endpoints and Codex binary paths. Requires `OPENAI_API_KEY`.
GitHub Copilot CLI	`base`	Full coverage	Pre-installed. Works out of the box. Requires `GITHUB_TOKEN` or `COPILOT_GITHUB_TOKEN`.
OpenClaw	`openclaw`	Bundled	Agent orchestration layer. Launch with `openshell sandbox create --from openclaw`.
Ollama	`ollama`	Bundled	Run cloud and local models. Includes Claude Code, Codex, and OpenCode. Launch with `openshell sandbox create --from ollama`.

OpenShell

Isolated Sandbox and Governance for Safe, Productive Claws

OFF

ON

Anywhere Developers Are

OpenClaw

+

NVIDIA OpenShell

Cloud

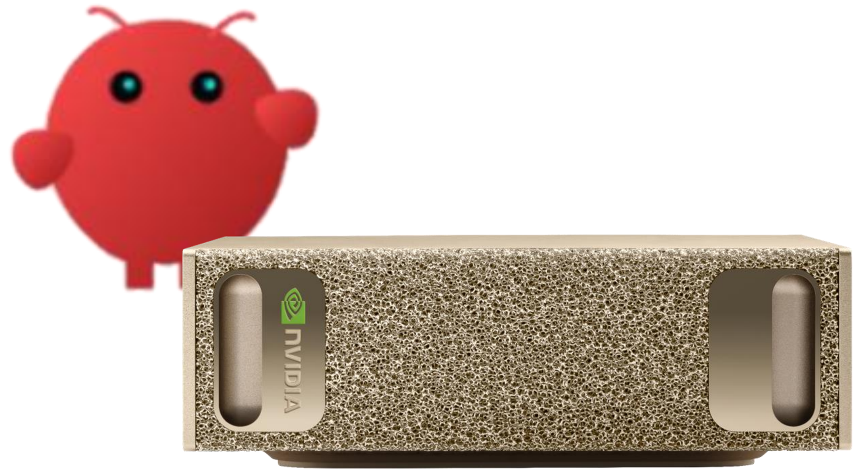
Data Center

Edge

Workstation

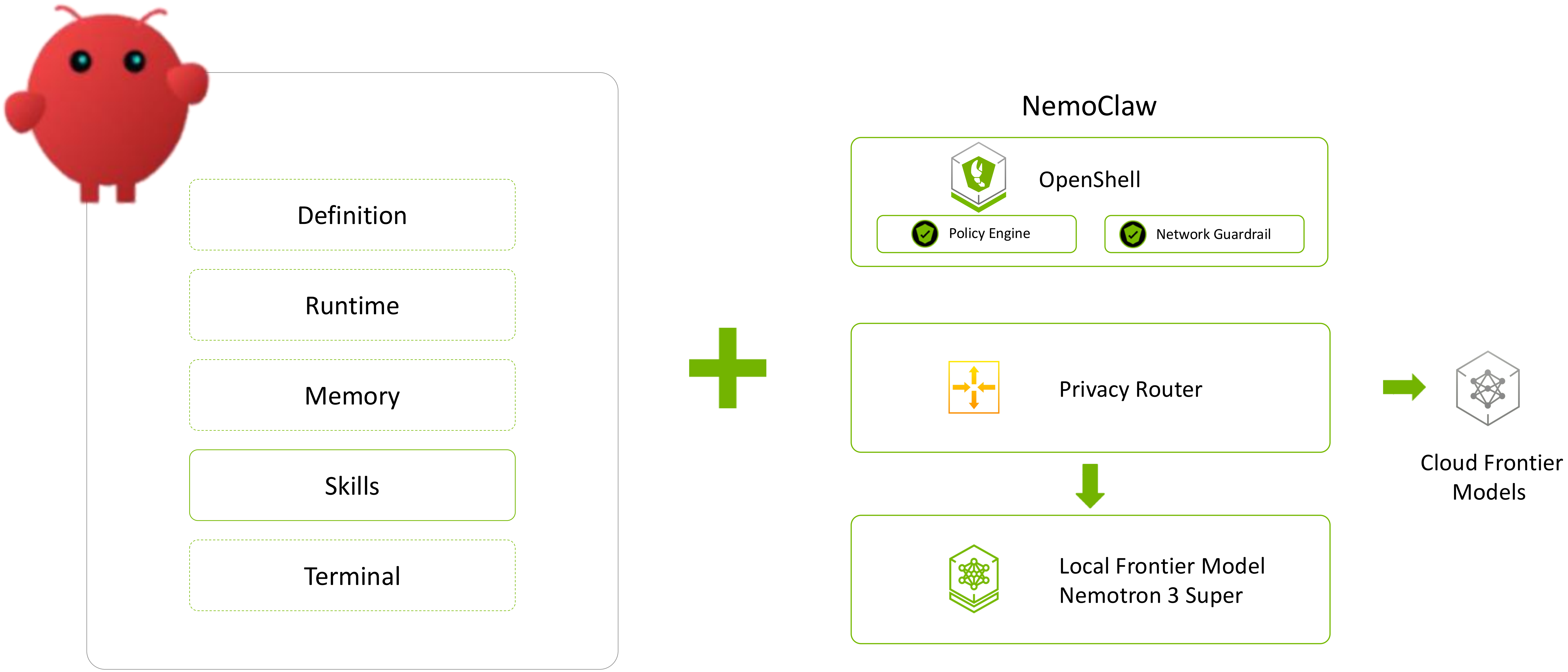
NVIDIA DGX Spark and DGX Station to Build Safer Autonomous, Self-Evolving Agents

DGX Spark



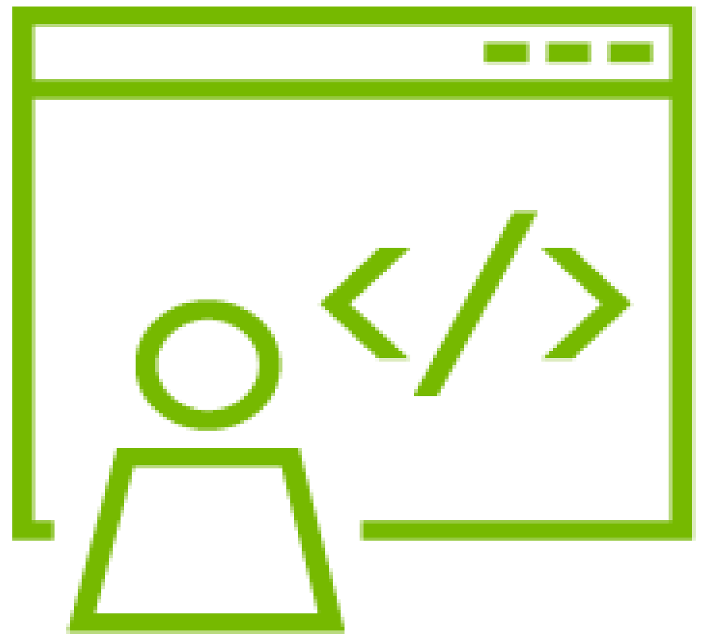
- Desktop AI Computer for Everyone
- 200B Parameter Models
- Power efficient for always on agents

NVIDIA NemoClaw for OpenClaw

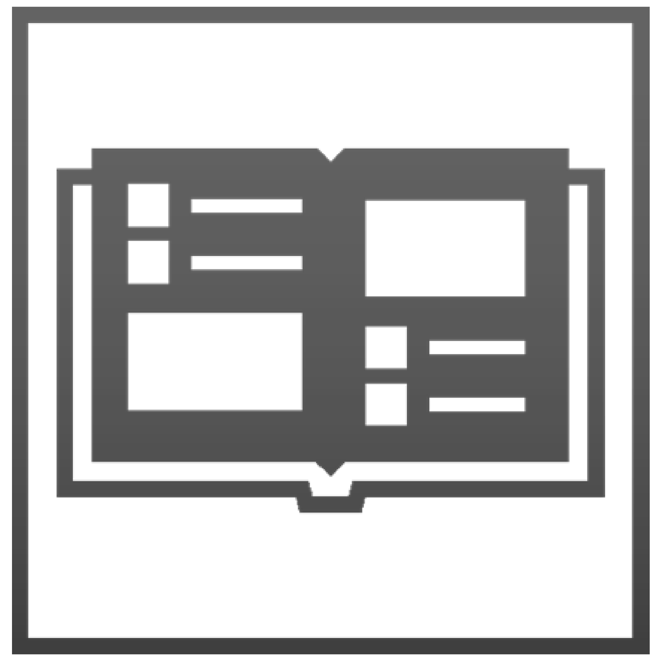


NVIDIA Agent Toolkit

Get Started with NVIDIA OpenShell



github.com/NVIDIA/OpenShell



docs.nvidia.com/openshell

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